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2.1.1 Design Properties

Import deck loads and functions. |

Table 1: Load values from rv102-loads.py (rv_stor/v102-2.csv)

variable	value	[value]	description
D_1	2.00 plf	0.00 klf	2x6 planks DL
D_2	2.60 plf	0.00 klf	2x8 joists DL
D_3	2.90 plf	0.00 klf	4x4 posts and struts
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
LL_1	40.00 psf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 psf	0.96 kPA	ASCE7-05 HL

Table 2: Function Import (rvsrc/scripts/sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL
bending_stress_udl(ln, w, b, d)	Maximum bending stress in a simply supported beam with UDL.
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.

2.1.2 Deck Design Summary

Design properties .

```
Function Arguments Dictionary : beam1 (units: inch, pounds)
=================================================================
ln_1 = 4*12.  # beam span
w_1 = 45*.5/12  # uniform linear load
b_1 = 5.5  # beam width
d_1 = 1.5  # beam depth
E_1 = 1.5*(10**6)  # modulus of elasticity
F_b = 1000 # allowable bending stress
C_D = 1.0   # load duration factor
C_M = 0.85 # wet service factor
C_F = 1.0  # size factor
C_t = 1.0  # temperature factor
C_i = 0.8  # incising factor
C_r = 1.0  # repetitive member factor
C_c = 1.0  # curvature factor
C_L = 1.0  # beam stability factor
C_b = 1.0  # bearing area factor
deflect_limit = 240.0 # max allowable deflection ln_1/deflect_limit
=================================================================
```

Design Results

Eq. 1: Check Deck Beam

```
nds_beam_check | units: inch, pounds

     Beam Check Results:
     ===============================
     total UDL: 1.88
     fb: 261.82
     Fb_prime: 680.00
     E_prime: 1275000.00
     stress_ratio: 0.39
     deflection: 0.07
     deflection_ratio:  0.20
```